Literature Status: Johnson–Odell Elastic Space Problem

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Status

This is a lightweight *literature already answered* packet for arXiv:math/0310023. It records that a source problem in Johnson–Odell is explicitly answered by the later paper arXiv:1502.03791.

Original Problem

In *The Diameter of the Isomorphism Class of a Banach Space*, Johnson and Odell define an elastic Banach space and ask in Problem 7:

> Let X be elastic (and separable, say). Does $C[0,1]$ embed into X?

The same paper proves the related main theorem that every separable infinite-dimensional Banach space has infinite Banach–Mazur isomorphism class diameter, and proves as an intermediate result that a separable infinite-dimensional elastic space contains c_0. Problem 7 asks for the stronger $C[0,1]$ conclusion.

Supporting Answer

Alspach and Sari, *Separable Elastic Banach spaces are universal*, arXiv:1502.03791, explicitly state that Johnson and Odell conjectured that the separable elastic spaces are exactly those containing an isomorphic copy of $C[0,1]$, and write that they prove this conjecture.

Their Theorem 1 states:

> Let X be a separable infinite dimensional elastic Banach space. Then $C[0,1]$ is isomorphic to a subspace of X.

This is an affirmative answer to the separable version of Johnson–Odell Problem 7.

Scope

The packet records only the separable elastic-space problem asked in Problem 7. It does not claim to settle any nonseparable extension of that problem or any nonseparable version of the Banach–Mazur diameter question.

References

- W. B. Johnson and E. Odell, *The Diameter of the Isomorphism Class of a Banach Space*, arXiv:math/0310023.

- Dale E. Alspach and Bunyamin Sari, *Separable Elastic Banach spaces are universal*, arXiv:1502.03791.